



Ain Shams University

Faculty of Engineering

Computer Engineering and Systems Program

CSE334s (UG2018) — Database Systems

Project Report – Hotel Database System - Phase II

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Team (19)

GitHub Repository

<https://github.com/Mathew-Aziz/Hotel-Database-System>

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1. Introduction

The Hotel Management System is a desktop application developed as part of the Database Systems course (CSE334s) at Ain Shams University. The project demonstrates the practical integration of a relational database with a graphical user interface, covering the full software development cycle from schema design to a working application.

The system is built to simulate the core daily operations of a hotel, allowing staff to manage guests, rooms, bookings, services, and personnel through a structured and user-friendly interface. Rather than relying on manual records or disconnected spreadsheets, the application centralizes all hotel data in a SQL Server database and exposes it through five dedicated WinForms screens, each responsible for a specific area of hotel management.

The backend database was designed with referential integrity in mind, enforcing foreign key relationships, check constraints, and cascade rules to ensure data consistency at all times. On the application side, all database interactions are handled through ADO.NET using parameterized queries, which prevents SQL injection and ensures safe, reliable communication between the interface and the database.

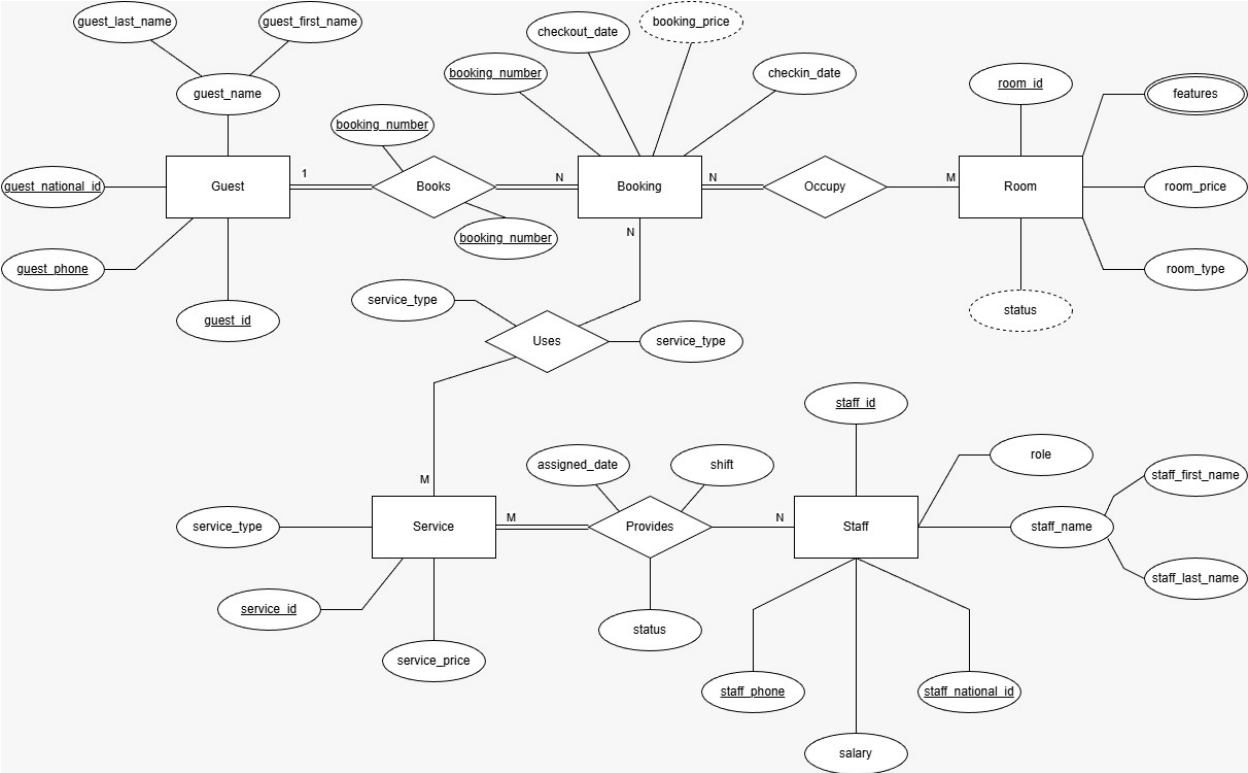
The project was divided into two phases. Phase I focused on entity-relationship modeling and schema design. Phase II, documented in this report, covers the implementation of the full database script including seed data, and the development of the five-form WinForms application with complete CRUD functionality across all major entities.

2. Theme

Hotel Management System — a Windows desktop application that helps hotel staff manage daily operations, including guest registration, room reservations, service tracking, and staff assignments. The system allows full data manipulation (Create, Read, Update, Delete) over five main entities, with a user-friendly graphical interface built with WinForms and connected to a SQL Server database.

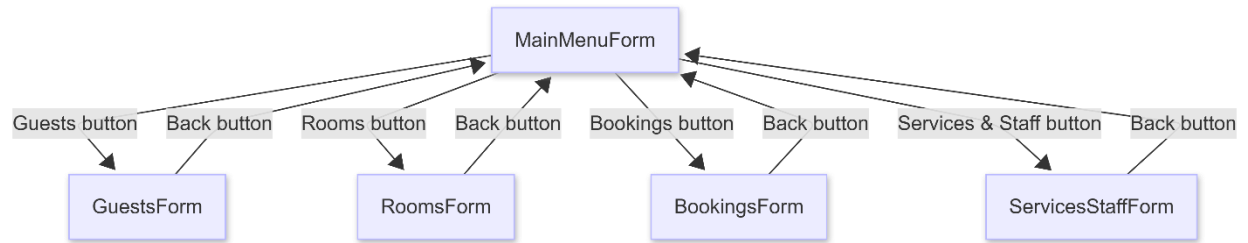
3. ERD Diagram

The Entity-Relationship Diagram below illustrates the full schema of the Hotel Management System database, showing all entities, attributes, and relationships.



4. Navigation Diagram

The navigation follows a simple hub-and-spoke model: the MainMenuForm stays visible and opens any child form as a new window. Each child form has a Back button that closes itself and returns to the main menu.



5. The 5 Forms and Their Functionalities

5.1 MainMenuForm

- Purpose: Navigation hub.
- Contains four buttons that open the other forms.
- No database interaction.

5.2 GuestsForm

- Purpose: Manage guest profiles.
 - **Select:** View all guests in a DataGridView.
 - **Insert:** Add new guest (national ID, first name, last name, phone).
 - **Update:** Modify the selected guest.
 - **Delete:** Remove a guest (shows a friendly error if the guest has existing bookings).

5.3 RoomsForm

- Purpose: Manage hotel rooms.
 - **Select:** List rooms with type and price.
 - **Insert:** Add a new room (type, price per night).
 - **Update:** Change room type or price.
 - **Delete:** Remove a room (blocked if the room is assigned to any booking).

5.4 BookingsForm

- Purpose: Manage reservations (core form).
 - **Select:** Display all bookings (guest name, dates, booking status, payment status).
 - **Insert:** Create a new booking (guest, dates, statuses).
 - **Update:** Modify dates, statuses, or guest.
 - **Delete:** Remove a booking — linked rooms and services are automatically removed via cascade constraints.
 - **Room management (Occupies):** Add or remove rooms to/from a booking — enforces at least one room per booking.
 - **Total price:** Calculates room cost (rooms x nights) and displays it dynamically.

5.5 ServicesStaffForm

Manages services, staff, service usage per booking (Uses), and staff-service assignments (Provides) — all in one scrollable window without tabs.

Services section (dbo.SERVICE)

- **Select:** View all services.
- **Insert:** Add a new service (type, positive price).

- **Update:** Modify service type or price.
- **Delete:** Remove a service (fails if used in any assignment or booking usage).

Staff section (dbo.Staff)

- **Select:** View all staff members.
- **Insert:** Add new staff (first name, last name, role, phone, national ID, salary ≥ 0).
- **Update:** Modify any staff field.
- **Delete:** Remove a staff member (fails if the staff has service assignments).

Uses section — Services per Booking

- **Select:** Shows all service usage records (booking, service, date, quantity, line total).
- **Insert:** Select a booking, a service, set date and positive quantity, then add.
- **Update:** Select a usage row and change quantity.
- **Delete:** Remove a service from a booking.

Provides section — Staff-Service Assignments

- **Select:** Shows all assignments (staff name, service type, assigned date, shift, status).
- **Insert:** Select a staff member, a service, set shift and status, then add.
- **Update:** Select an assignment row and change shift or status.
- **Delete:** Remove an assignment.

All four sections support full CRUD (Create, Read, Update, Delete). Parameterised queries prevent SQL injection, and comprehensive validation and error handling (e.g. foreign-key violations, duplicate keys) provide clear user feedback.

6. Technologies Used

Technology	Details
Frontend	Windows Forms (.NET 8 / .NET Framework)
Language	C#
Database	Microsoft SQL Server (localhost and SQL Express)
Data Access	ADO.NET with Microsoft.Data.SqlClient (parameterised queries)
Database Script	HotelDB.sql — creates tables, constraints, foreign keys (with cascade), seeds 5+ rows per table

7. Conclusion

This project successfully delivered a fully functional Hotel Management System that satisfies all requirements outlined for Phase II of the Database Systems course. The application demonstrates a complete and working integration between a relational SQL Server database and a Windows Forms desktop interface, covering every core database operation — Select, Insert, Update, and Delete — across all five forms.

The database schema was designed with careful attention to data integrity, incorporating primary and foreign key constraints, check constraints on critical fields such as price, salary, and quantity, and cascade delete rules where appropriate to maintain consistency when parent records are removed. The seed data provided in the SQL script ensures the application can be demonstrated immediately on any fresh database without manual data entry.

On the application side, the team implemented a clean navigation model, thorough input validation, and user-friendly error handling that translates database constraint violations into readable messages rather than exposing raw exceptions to the user. All queries use parameterized inputs, ensuring the application is protected against SQL injection throughout.

Through this project, the team gained practical experience in relational database design, the use of ADO.NET for data access in C#, and the challenges of coordinating between a database schema and a graphical interface in a real multi-table system. The iterative process of identifying missing functionality, fixing validation gaps, and aligning the GUI with the finalized schema reflects the kind of integration work required in real software development.

The final deliverable is a stable, demonstrable application that meets all submission requirements and serves as a solid foundation for any further development, such as reporting features, user authentication, or migration to a web-based interface.